

CSE 20 Winter 2026

Exam Practice Solutions

- 1. Modeling, Sets and Functions, Algorithms** In machine learning, clustering can be used to group similar data for prediction and recommendation. For example, each Netflix user's viewing history can be represented as a n -tuple indicating their preferences about movies in the database, where n is the number of movies in the database. Each element in the n -tuple indicate the user's rating of the corresponding movie: 1 indicates the person liked the movie, -1 that they didn't, and 0 that they didn't rate it one way or another.

Consider the following algorithm for determining if a user's viewing history represents strong opinions on many movies.

Determine if user's ratings tuple encode strong opinions

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1 procedure opinion(( $r_1, \dots, r_n$ ): a  $n$ -tuple of ratings;  $c$ : a nonnegative integer)
2    $sum := 0$ 
3   for  $i := 1$  to  $n$ 
4     if  $r_i \neq 0$ 
5        $sum := sum + 1$ 
6   return  $sum \geq c$ 
```

1. When $n = 3$, describe the set of all n -tuples representing user ratings
- (a) By the roster method.

There are 27 3-tuples whose elements are each -1 , 0 , or 1 . We can list them as follows.

$$\{(-1, -1, -1), (-1, -1, 0), (-1, -1, 1), (-1, 0, -1), (-1, 0, 0), (-1, 0, 1), \\ (-1, 1, -1), (-1, 1, 0), (-1, 1, 1), (0, -1, -1), (0, -1, 0), (0, -1, 1), \\ (0, 0, -1), (0, 0, 0), (0, 0, 1), (0, 1, -1), (0, 1, 0), (0, 1, 1), (1, -1, -1), \\ (1, -1, 0), (1, -1, 1), (1, 0, -1), (1, 0, 0), (1, 0, 1), (1, 1, -1), (1, 1, 0), (1, 1, 1)\}$$

- (b) With set builder notation.

Using set builder notation, we can specify the constraints on each component of the 3-tuple

$$\{(x, y, z) \mid x \in \{-1, 0, 1\} \text{ and } y \in \{-1, 0, 1\} \text{ and } z \in \{-1, 0, 1\}\}$$

- (c) Using a recursive definition.

Let's call the set of ratings of three movies R_3 . We can define this set recursively by listing each of the finitely many tuples in R_3 in the basis step and then defining the recursive step to be: when $u \in R_3$, so is u .

Note that this recursive definition is somewhat contrived: the recursive step isn't doing anything helpful.

2. What are the possible return (output) values of this algorithm?

Since the algorithm returns the result of a comparison, the possible values are True or False.

3. **Trace** the computation of $opinion((-1, 0, 0, 0, 1, 1, -1), 2)$.

The computation of $opinion((-1, 0, 0, 0, 1, 1, -1), 2)$ first (in line 2) initializes the variable sum to 0 and then steps through each component of the tuple in the for loop in line 3, checking (in line 4) if the value is zero or not. The first component of $(-1, 0, 0, 0, 1, 1, -1)$ is nonzero so line 5 increments sum , which now has value 1. When the for loop checks the components at positions 2,3,4, the sum variable is not updated. When the for loop gets to each of positions 5,6,7, the nonzero value of the elements at these positions means the if branch of the condition is followed and the sum variable is incremented three times, to get to the value 4. In line 6, the comparison $sum \geq c$ evaluates to $4 \geq 2$, which is True. Thus, $opinion((-1, 0, 0, 0, 1, 1, -1), 2)$ returns True.

4. Give an example of a nonnegative integer c so that, no matter which n -tuple $(r_1, \dots, r_n)$ we consider, $opinion((r_1, \dots, r_n), c)$ will have the same value. What value is it?

The nonnegative integer $c = 0$ is an example. The value of sum is initialized to 0 in line 2. The values of the components n -tuple $(r_1, \dots, r_n)$ determine whether the value of sum ever increases in the for loop lines 3-5; the value can never decrease. Thus, no matter the choice of n -tuple, the comparison $sum \geq 0$ will evaluate to True.

2. Sets and Functions

1. Give a recursive definition for the set $\mathbb{N}$.

The set of nonnegative integers can be defined recursively as follows:

Basis step: $0 \in \mathbb{N}$

Recursive step: If $x \in \mathbb{N}$ then $x + 1 \in \mathbb{N}$.

2. Use the recursive definition from part (a) to give a recursive definition for the function with domain $\mathbb{N}$, codomain $\mathbb{N}$ which computes, for input i , the sum of the first i powers of 2. For example, on input 0, the function evaluates to 2^0 , namely to 1. On input 1, the function evaluates to $2^0 + 2^1$, namely 3. On input 2, the function evaluates to $2^0 + 2^1 + 2^2$, namely 7.

Let's call this function $sumPow$. Its recursive definition will mirror the recursive definition of its domain, from part (a).

$sumPow : \mathbb{N} \rightarrow \mathbb{N}$

Basis step: $sumPow(0) = 1$.

Recursive step: If $x \in \mathbb{N}$ then $sumPow(x+1) = sumPow(x) + 2^{x+1}$.

In this function definition, we assumed that we have access to a definition of the power function to compute powers of two. In fact, we can define this function recursively as well:

$powTwo : \mathbb{N} \rightarrow \mathbb{N}$

Basis step: $powTwo(0) = 1$.

Recursive step: If $x \in \mathbb{N}$ then $powTwo(x+1) = 2powTwo(x) = powTwo(x) + powTwo(x)$.

3. Base expansions, Multiple representations

1. Compute the ternary (base 3) expansion of 28.

We can express 28 in terms of powers of 3 as

$$28 = 1 \cdot 27 + 1 = 1 \cdot 3^3 + 0 \cdot 3^2 + 0 \cdot 3^1 + 1 \cdot 3^0 = (1001)_3.$$

Therefore, by definition of base 3 expansion, $28 = (1001)_3$.

2. Compute the product of $(6A)_{16}$ and $(11)_{16}$, without converting either number to another base.

We use the usual algorithm for multiplication, except using hexadecimal symbols.

$$\begin{array}{r} 6A \\ \times 11 \\ \hline 6A \\ +6A0 \\ \hline 70A \end{array}$$

Thus, the product of $(6A)_{16}$ and $(11)_{16}$ is $(70A)_{16}$.

3. Confirm your answer for part (b) by converting $(6A)_{16}$ and $(11)_{16}$ to decimal, multiplying them, and converting the product to base 16.

Converting:

$$(6A)_{16} = 6 \cdot 16^1 + 10 \cdot 16^0 = 96 + 10 = 106$$

$$(11)_{16} = 1 \cdot 16^1 + 1 \cdot 16^0 = 16 + 1 = 17$$

Multiplying in decimal:

$$\begin{array}{r} 106 \\ \times 17 \\ \hline 742 \\ +1060 \\ \hline 1802 \end{array}$$

Converting back to hexadecimal: by long division,

$$1802 = 112 \cdot 16 + 10$$

$$112 = 7 \cdot 16 + 0$$

$$7 = 0 \cdot 16 + 7$$

Thus, $1802 = (70A)_{16}$, agreeing with part (b).

4. How many bits will there be in the binary (base 2) expansion of 2020? Can you compute this without fully converting 2020 to base 2?

The leading coefficient will need to be in the column corresponding to the highest power of 2 less than or equal to 2020. Listing the powers of 2:

Exponent	0	1	2	3	4	5	6	7	8	9	10	11
Power	1	2	4	8	16	32	64	128	256	512	1024	2048

we see that the binary expansion of 2020 will be of the form $(1a_9 \cdots a_0)_2$ and will therefore have 11 bits.

5. Give an example of a number that can be represented in base 2 fixed-width 3, but not in base 2 expansion.

The number 0 can't be represented in binary (base 2) but is represented in base 2 fixed-width 3 $(000)_{2,3}$.

6. Give an example of a number that can be represented in base 2 expansion, but not in base 2, fixed-width 3 expansion.

The number 8 can be represented in base 2 as $(1000)_2$ but can't be represented in binary fixed-width 3 because it requires four bits.

7. Give the representation of -7 in sign-magnitude width 3 and 2s complement width 3. Then do the same for width 4.

The number -7 can't be represented in sign magnitude width 3 or in 2s complement width 3:

- The least number that can be represented in sign magnitude width 3 has representation $[111]_{s,3}$ and is -3 .
- The least number that can be represented in 2s complement width 3 has representation $[100]_{2c,3}$ and is -4 .

With width 4 we can use the following representations:

- In sign magnitude width 4, -7 has representation $[1111]_{s,4}$ because the leftmost bit is the sign bit and the magnitude of -7 has binary fixed width 3 representation $7 = (111)_{2,3}$.
- In 2s complement width 4, -7 has representation $[1001]_{2c,4}$ because the leftmost bit is the sign bit and rightmost bits need to represent $2^{4-1} - 7 = 1$ in binary fixed width 3, $(001)_{2,3}$. Alternatively, we calculate that $[1001]_{2c,4} = 1 \cdot (-2)^3 + 0 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 = -8 + 1 = -7$.

8. Give the representation of 10.5625 in fixed-width base-2 expansion with integer width 4 and fractional width 9.

We need to write 10.5625 as

$$(a_3a_2a_1a_0.c_1c_2c_3c_4c_5c_6c_7c_8c_9)_{2,4,9}$$

For the integer part: $10 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 = (1010)_2$

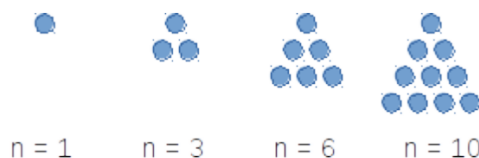
For the fractional part: $0.5625 = 1 \cdot 2^{-1} + 0 \cdot 2^{-2} + 0 \cdot 2^{-3} + 1 \cdot 2^{-4}$ where

Exponent	-1	-2	-3	-4	-5	-6	-7
Power	0.5	0.25	0.125	0.0625	0.03125	0.015625	0.0078125

Thus,

$$10.5625 = (1010.100100000)_{2,4,9}$$

- 4. Circuits** A triangular number (or triangle number) counts the objects that can form an equilateral triangle, as in the diagram below. The n^{th} triangular number is the sum of the first n integers, as shown in the following figure illustrating the first four triangular numbers (what is the fifth one?):



Design a circuit that takes four inputs x_0, x_1, x_2, x_3 and outputs True (T or 1) if the integer value $(x_3x_2x_1x_0)_{2,4}$ is a triangular number, and False (F or 0) otherwise. You

may assume that 0 is not a triangular number. (*Credit: UBC Department of Computer Science*)

Using the definition, we can write a table of values for the circuit we need to build, since the output will be one when $(x_3x_2x_1x_0)_{2,4}$ will be one of 1, 3, 6, 10, 15.

x_3	x_2	x_1	x_0	Output
1	1	1	1	1
1	1	1	0	0
1	1	0	1	0
1	1	0	0	0
1	0	1	1	0
1	0	1	0	1
1	0	0	1	0
1	0	0	0	0
0	1	1	1	0
0	1	1	0	1
0	1	0	1	0
0	1	0	0	0
0	0	1	1	1
0	0	1	0	0
0	0	0	1	1
0	0	0	0	0

The output can be described using a compound proposition in DNF with five clauses (because there are five rows with output 1):

$$(x_3 \wedge x_2 \wedge x_1 \wedge x_0) \vee (x_3 \wedge \neg x_2 \wedge x_1 \wedge \neg x_0) \vee (\neg x_3 \wedge x_2 \wedge x_1 \wedge \neg x_0) \\ \vee (\neg x_3 \wedge \neg x_2 \wedge x_1 \wedge x_0) \vee (\neg x_3 \wedge \neg x_2 \wedge \neg x_1 \wedge x_0)$$

We can use this DNF to draw a circuit that takes the four inputs and whose output is described.

Alternatively, a logically equivalent compound proposition is

$$output = [x_1 \wedge (x_0 \oplus x_2) \oplus x_3] \vee [\neg x_3 \wedge \neg x_2 \wedge \neg x_1 \wedge x_0]$$

Using this compound proposition, we can draw a circuit with fewer gates than the one described above.

Note: we didn't include picture of circuits in these sample solutions; please come to office hours or post to Piazza if you'd like to get feedback on a circuit you draw.

5. Circuits, Compound Propositions, Logical Equivalence

1. Draw a logic circuit that uses **exactly three** gates and is logically equivalent to

$$q \leftrightarrow (p \wedge r)$$

You may (only) use AND, OR, NOT, and XOR gates.

Since XOR gates are available, we can use a convenient relationship between XOR and $\leftrightarrow$:

$$q \leftrightarrow (p \wedge r) \equiv \neg(q \oplus (p \wedge r))$$

Implementing this compound proposition as a circuit, we will use one XOR gate, one AND gate, and one NOT gate, as required.

Note: we didn't include picture of circuits in these sample solutions; please come to office hours or post to Piazza if you'd like to get feedback on a circuit you draw.

2. Write a compound proposition which is logically equivalent to

$$(p \oplus q) \leftrightarrow r$$

You may only use the logical operators negation ($\neg$), conjunction ($\wedge$), and disjunction ($\vee$).

We apply several logical equivalences in turn: first, we can rewrite $\leftrightarrow$ in terms of $\wedge$ and $\rightarrow$:

$$(p \oplus q) \leftrightarrow r \equiv ((p \oplus q) \rightarrow r) \wedge (r \rightarrow (p \oplus q))$$

Now, we replace $\rightarrow$ by its equivalent disjunctive form: $HYP \rightarrow CONC \equiv (\neg HYP) \vee CONC$. Thus,

$$((p \oplus q) \rightarrow r) \wedge (r \rightarrow (p \oplus q)) \equiv (\neg(p \oplus q) \vee r) \wedge (\neg r \vee (p \oplus q))$$

Last, we replace the XOR with equivalent compound propositions in terms of the other connectives: the XOR is true means one of its inputs is T and the other F; the XOR is false means its inputs are both T or both F.

$$\begin{aligned} & (\neg(p \oplus q) \vee r) \wedge (\neg r \vee (p \oplus q)) \equiv \\ & (((p \wedge q) \vee (\neg p \wedge \neg q)) \vee r) \wedge (\neg r \vee ((p \wedge \neg q) \vee (\neg p \wedge q))) \end{aligned}$$

3. Find a compound proposition that is in DNF (disjunctive normal form) and is logically equivalent to

$$(p \vee q \vee \neg r) \wedge (p \vee \neg q \vee r) \wedge (\neg p \vee q \vee r)$$

The given compound proposition is in CNF and indicates that the rows in the truth table with $p = F, q = F, r = T$, $p = F, q = T, r = F$, and $p = T, q = F, r = F$ are set to F. Thus, the truth table is

p	q	r	Output
T	T	T	T
T	T	F	T
T	F	T	T
T	F	F	F
F	T	T	T
F	T	F	F
F	F	T	F
F	F	F	T

To find a logically equivalence compound proposition in DNF, we represent the disjunction of landing in each of the rows that evaluate to T : the rows where $p = T, q = T, r = T$, or $p = T, q = T, r = F$, or $p = T, q = F, r = T$, or $p = F, q = T, r = T$, or $p = F, q = F, r = F$.

$$(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \\ \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r)$$

6. Translating Propositional Logic

p is “The display is 13.3-inch”

r is “There is at least 128GB of flash storage”

u is “There is at least 512GB of flash storage”

q is “The processor is 2.2 GHz”

s is “There is at least 256GB of flash storage”

1. Are the statements

$$p \rightarrow (r \vee s \vee u) \quad , \quad q \rightarrow (s \vee u) \quad , \quad p \leftrightarrow q \quad , \quad \neg u$$

consistent? If so, translate to English a possible assignment of truth values to the input propositions that makes all four statements true simultaneously.

Yes, these statements are consistent. Consider the assignment of truth values

$$p = T \quad q = T \quad r = T \quad s = T \quad u = F$$

We evaluate the four compound propositions at this assignment to confirm that it makes all of them true simultaneously:

$$p \rightarrow (r \vee s \vee u) = T \rightarrow (T \vee T \vee F) = T \rightarrow T = T$$

$$q \rightarrow (s \vee u) = T \rightarrow (T \vee F) = T \rightarrow T = T$$

$$p \leftrightarrow q = T \leftrightarrow T = T$$

$$\neg u = \neg F = T$$

Translating to English, we get “The display is 13.3-inch, the processor is 2.2GHz, there is at least 128GB of flash storage, there is at least 256GB of flash storage, but there is not at least 512GB of flash storage.”

2. Consider this statement in English:

It’s not the case that both the display is 13.3-inch and the processor is 2.2 GHz.

Determine whether each of the compound propositions below is equivalent to the **negation** of that statement, and justify your answers using either truth tables or other equivalences.

Possible compound propositions:

- $\neg p \vee \neg q$
- $\neg(p \rightarrow \neg q)$
- $\neg(p \wedge q)$
- $(\neg p \leftrightarrow \neg q) \wedge p$

The sentence can be translated to $\neg(p \wedge q)$. Its negation is then

$$\neg\neg(p \wedge q) \equiv (p \wedge q) \equiv \neg(p \rightarrow \neg q)$$

because $A \wedge \neg B \equiv \neg(A \rightarrow B)$ so $p \wedge q \equiv \neg(A \rightarrow \neg B)$.

This is also equivalent to $(\neg p \leftrightarrow \neg q) \wedge p$, as we can see from the last two columns of the the truth table below

p	q	$(\neg p \leftrightarrow \neg q)$	$(\neg p \leftrightarrow \neg q) \wedge p$	$p \wedge q$
T	T	T	T	T
T	F	F	F	F
F	T	F	F	F
F	F	T	F	F

3. Consider the compound proposition

$$(p \wedge q) \rightarrow (r \vee s \vee u)$$

Express the **contrapositive** of this conditional as a compound proposition.

Then, give an assignment of truth values to each of the input propositional variables for which the original compound proposition is True but its **converse** is False.

The contrapositive of this conditional is

$$\neg(r \vee s \vee u) \rightarrow \neg(p \wedge q)$$

The converse of the original conditional is

$$(r \vee s \vee u) \rightarrow (p \wedge q)$$

Consider the assignment of truth values

$$p = T \quad q = F \quad r = T \quad s = F \quad u = F$$

We evaluate the original conditional and its converse with this assignment of truth values:

$$\text{Original : } (p \wedge q) \rightarrow (r \vee s \vee u) = (T \wedge F) \rightarrow (T \vee F \vee F) = F \rightarrow T = T$$

$$\text{Converse : } (r \vee s \vee u) \rightarrow (p \wedge q) = (T \vee F \vee F) \rightarrow (T \wedge F) = T \rightarrow F = F$$

7. Evaluating predicates, Evaluating nested predicates

- Over the domain $\{1, 2, 3, 4, 5\}$ give an example of predicates $P(x), Q(x)$ which demonstrate that

$$(\forall x P(x)) \vee (\forall x Q(x)) \quad \text{is not logically equivalent to} \quad \forall x (P(x) \vee Q(x))$$

We will define predicates which make one of the statements False and the other True. Consider

x	$P(x)$	$Q(x)$
1	T	F
2	F	T
3	T	F
4	F	T
5	T	F

Evaluating the statements: $\forall x (P(x) \vee Q(x))$ is True because each row in the table has at least one of $P(x), Q(x)$ evaluate to True. On the other hand, neither $\forall x P(x)$ nor $\forall x Q(x)$ is True, because each predicate at least one domain element where it evaluates to False (for counterexamples: $P(2)$ is False and $Q(1)$ is False). Since $F \vee F = F$, $(\forall x P(x)) \vee (\forall x Q(x))$ is False. Since there is some definition of the predicates P and Q where the two statements

$$(\forall x P(x)) \vee (\forall x Q(x)) \quad , \quad \forall x (P(x) \vee Q(x))$$

evaluate to different truth values, they are not logically equivalent

2. Over the domain $\{0, 1, 2\}$, give an example of predicates $P(x), Q(x)$ for which all of these statements are true:

$$\forall x(P(x) \rightarrow Q(x))$$

$$\exists x P(x)$$

$$\exists x \neg P(x)$$

$$\exists x \neg Q(x)$$

We will define predicates using their tables of values:

x	$P(x)$	$Q(x)$
0	T	T
1	F	T
2	F	F

We now prove that each of the statements is True:

- To prove the universal statement $\forall x(P(x) \rightarrow Q(x))$, we evaluate the conditional at each element in the domain: At 0, $P(0) \rightarrow Q(0) = T \rightarrow T = T$; at 1, $P(1) \rightarrow Q(1) = F \rightarrow T = T$; at 2, $P(2) \rightarrow Q(2) = F \rightarrow F = T$. Since the body of the quantification is True at all element in the domain, $\forall x(P(x) \rightarrow Q(x))$ is True.
- To prove the existential statement $\exists x P(x)$, we need a witness in the domain where $P(x)$ evaluates to True. The witness is 0, since the first row in the definition table gives $P(0) = T$.
- To prove the existential statement $\exists x \neg P(x)$, we need a witness in the domain where $\neg P(x)$ evaluates to True. The witness is 1, since the second row in the definition table gives $P(1) = F$ so $\neg P(1) = T$.
- To prove the existential statement $\exists x \neg Q(x)$, we need a witness in the domain where $\neg Q(x)$ evaluates to True. The witness is 2, since the third row in the definition table gives $Q(2) = F$ so $\neg Q(2) = T$.

8. Evaluating predicates, Evaluating nested predicates Recall that S is defined as the set of all RNA strands, where each strand is a nonempty string made of the bases in $B = \{\mathbf{A}, \mathbf{U}, \mathbf{G}, \mathbf{C}\}$. Recall the definition of the following predicates: $L_{\mathbf{A}}$ with domain S is defined recursively by:

Basis step: $L_{\mathbf{A}}(\mathbf{A}) = T, L_{\mathbf{A}}(\mathbf{C}) = L_{\mathbf{A}}(\mathbf{G}) = L_{\mathbf{A}}(\mathbf{U}) = F$

Recursive step: If $s \in S$ and $b \in B$, then $L_{\mathbf{A}}(sb) = L_{\mathbf{A}}(s)$

$P_{\mathbf{AUC}}$ with domain S is defined as the predicate whose truth set is the collection of RNA strands where the string **AUC** is a substring (appears inside s , in order and consecutively)

L with domain $S \times \mathbb{Z}^+$ is defined by, for $s \in S$ and $n \in \mathbb{Z}^+$,

$$L(s, n) = \begin{cases} T & \text{if } \text{rlen}(s) = n \\ F & \text{otherwise} \end{cases}$$

1. Give a witness for $\exists s_1 \exists s_2 (L(s_1, 4) \wedge L(s_2, 4) \wedge \neg(s_1 = s_2))$ where S is the set of RNA strands, $L(s, n)$ is a predicate with domain $S \times \mathbb{Z}^+$ that is true when s has length n

In English, the statement can be translated to “There are strands s_1, s_2 where each has length 4 and the strands are not equal.” A witness is $s_1 = \text{CCCC}$, $s_2 = \text{GGGG}$. Since each of s_1, s_2 is an element of S , s_1 has length 4, s_2 has length 4 and these two strings are not the same.

2. Give a counterexample that disproves $\forall i (\neg L(\text{ACU}, i))$

In English, the statement can be translated to “No positive integer is the length of the strand ACU.”
A counterexample that helps us disprove this statement is 3. This is a positive integer and is the length of ACU so $L(\text{ACU}, 3)$ is True, hence $\neg L(\text{ACU}, 3)$ is False.

3. Determine which of the following statements is True or False (you do not need to prove your answer).

(a) $\exists n \in \mathbb{Z}^+ \forall s \in S (P_{\text{AUC}}(s) \rightarrow \neg L(s, n))$

In English, this statement is saying that there is a positive integer that is not the length of every RNA strand with AUC. This is **true**, as we can see with the witness $n = 1$.

(b) $\exists n \in \mathbb{Z}^+ \forall s \in S (P_{\text{AUC}}(s) \wedge \neg L(s, n))$

In English, this statement is saying that there is a positive integer that is not the length of any RNA strand and that each RNA strand has AUC as a substring. This is **false**, because when we consider any positive integer, we’ll be able to find a counterexample strand that does not have AUC as a substring and so makes the conjunction false.

(c) $\forall s \in S \exists n \in \mathbb{Z}^+ (P_{\text{AUC}}(s) \vee \neg L(s, n))$

In English, this statement is saying that for each RNA strand we can find some positive integer so that at least one of “the strand has AUC as a substring” and “this number is not the length of this strand” is true. This is **true**, because given any RNA strand, we can pick a witness positive integer by choosing a number that is not its length.

(d) $\forall s \in S \exists n \in \mathbb{Z}^+ (L(s, n) \rightarrow \neg P_{\text{AUC}}(s))$

In English, this statement is saying that for each RNA strand, there is a witness positive integer which makes the conditional statement “if the RNA strand has this length then it does not have AUC as a substring” true. This is **true**, because given any RNA strand, we can pick a witness positive integer by choosing a number that is not its length which will make the hypothesis of the conditional statement false and therefore the conditional will be true.

9. Predicates and Quantifiers Express each of the following statements symbolically using quantifiers, variables, propositional connectives, and predicates (make sure to define the domain and the meaning of any predicates you introduce). Then, express its negation as a logically equivalent compound proposition without a $\neg$ in front. Decide whether the statement or its negation is true, and prove it.

- (a) If m is an integer, $m^2 + m + 1$ is odd.

Define the predicate $D(x)$ to mean “ x is odd” with domain $\mathbb{Z}$.

The sentence is then $\forall m \in \mathbb{Z}(D(m^2 + m + 1))$.

Its negation is $\neg \forall m \in \mathbb{Z}(D(m^2 + m + 1)) \equiv \exists m(\neg D(m^2 + m + 1))$.

The original statement is true.

Proof Towards universal generalization, let m be an arbitrary integer. By definition of odd, **to show**: there is an integer k such that $m^2 + m + 1 = 2k + 1$. Factoring the LHS, $m^2 + m + 1 = m(m + 1) + 1$. We proceed in proof by cases, since m is either even or odd.

- **Case 1 to show**: $(m \text{ is even }) \rightarrow D(m^2 + m + 1)$. In a direct proof of the conditional, assume m is even and choose c be an integer such that $m = 2c$. **To show**: there is an integer such that $m^2 + m + 1$ is twice that integer plus 1. To find a witness for this existential, we substitute $m = 2c$ and calculate, $m(m + 1) = 2c(2c + 1) = 2(2c^2 + c)$. Since $2c^2 + c \in \mathbb{Z}$ (by properties of integer addition and multiplication), it is the witness we need, because $2(2c^2 + c) + 1 = 2c(2c + 1) + 1 = m(m + 1) + 1 = m^2 + m + 1$.
- **Case 2 to show**: $(m \text{ is odd }) \rightarrow D(m^2 + m + 1)$. In a direct proof of the conditional, assume m is odd and choose c be an integer such that $m = 2c + 1$. **To show**: there is an integer such that $m^2 + m + 1$ is twice that integer plus 1. To find a witness for this existential, we substitute $m = 2c + 1$ and calculate, $m(m + 1) = (2c + 1)(2c + 2) = 2(2c + 1)(c + 1)$. Since $(2c + 1)(c + 1) \in \mathbb{Z}$ (by properties of integer addition and multiplication), it is the witness we need, because $2(2c + 1)(c + 1) + 1 = m(m + 1) + 1 = m^2 + m + 1$.

Alternate proof Let m be an integer. WTS that $m^2 + m + 1 \pmod{2} = 1$. Case 1: $m \pmod{2} = 0$. Then by modular arithmetic (Corollary 2 on page 242),

$$\begin{aligned}(m^2 + m + 1) \pmod{2} &= ((m \pmod{2})(m \pmod{2}) + (m \pmod{2}) + 1) \pmod{2} \\ &= (0 \cdot 0 + 0 + 1) \pmod{2} = 1\end{aligned}$$

as required. Case 2: $m \pmod{2} = 1$. Then by modular arithmetic (Corollary 2 on page 242),

$$\begin{aligned}(m^2 + m + 1) \pmod{2} &= ((m \pmod{2})(m \pmod{2}) + (m \pmod{2}) + 1) \pmod{2} \\ &= (1 \cdot 1 + 1 + 1) \pmod{2} = 3 \pmod{2} = 1\end{aligned}$$

as required.

- (b) For all integers n greater than 5, $2^n - 1$ is not prime.

Define the predicate: $P(x)$ to mean “ x is prime” with domain: $\mathbb{Z}$.

The sentence translates to $\forall n(n > 5 \rightarrow \neg P(2^n - 1))$.

Its negation is $\neg \forall n \in \mathbb{Z}(n > 5 \rightarrow \neg P(2^n - 1)) \equiv \exists n \in \mathbb{Z}(n > 5 \wedge P(2^n - 1))$.

The negation is true: to prove it, we need a witness integer greater than 5 where $2^n - 1$ is prime. Consider $n = 7$, an integer so in the domain. Calculating, $2^7 - 1 = 128 - 1 = 127$. This is a prime number because all integers greater than 1 and less than $\sqrt{127}$ (namely 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) are not factors of 127.

(c) If n and m are even integers, then $n - m$ is even.

Define the predicate $E(x)$ to mean “ x is even” with domain $\mathbb{Z}$. Note: this predicate could be expressed symbolically as $\exists k \in \mathbb{Z}(x = 2k)$.

The sentence translates to $\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} ((E(x) \wedge E(y)) \rightarrow E(x - y))$.

Its negation is

$$\neg \forall x \in \mathbb{Z} \forall y \in \mathbb{Z} ((E(x) \wedge E(y)) \rightarrow E(x - y)) \equiv \exists x \in \mathbb{Z} \exists y \in \mathbb{Z} (E(x) \wedge E(y) \wedge \neg E(x - y))$$

The original statement is true: Towards universal generalization let x, y be arbitrary integers and to show is $(E(x) \wedge E(y)) \rightarrow E(x - y)$. Assume, towards a direct proof of the conditional that x and y are both even. We WTS that $x - y$ is even. By definition of even, there are integers j, h such that $x = 2j$ and $y = 2h$. Then $x - y = (2j) - (2h) = 2(j - h)$. Choosing the witness $k = j - h$ (an integer), we have proved that $x - y$ is even.

10. Proof strategies Prove each of the following claims. Use only basic definitions and general proof strategies in your proofs; do not use any results proved in class / the textbook as lemmas. You may use basic properties of real numbers that we stated (without proof) in class, e.g. that the difference of two integers is an integer.

(a) The difference of any rational number and any irrational number is irrational.

Proof: Symbolically, to show is:

$$\forall x \in \mathbb{R} \forall y \in \mathbb{R} ((x \in \mathbb{Q} \wedge y \notin \mathbb{Q}) \rightarrow (x - y \notin \mathbb{Q}))$$

. Towards universal generalization, let x, y be arbitrary real numbers. Proceed in a proof by contradiction with

$$p = (x \in \mathbb{Q} \wedge y \notin \mathbb{Q}) \rightarrow (x - y \notin \mathbb{Q}) \quad r = y \in \mathbb{Q}$$

. To show: $\neg p \rightarrow (r \wedge \neg r)$. Assume $\neg p$, that is $x \in \mathbb{Q} \wedge y \notin \mathbb{Q} \wedge x - y \in \mathbb{Q}$. To show: $r \wedge \neg r$. By definition of conjunction, each conjunct is true and $y \notin \mathbb{Q}$ being true means $\neg(y \in \mathbb{Q})$ is true. Thus, $\neg r$ is true, and to show is: r . By definition of $\mathbb{Q}$, there are integers a, b, p, q with $b \neq 0$ and $q \neq 0$ such that $x = \frac{a}{b}, x - y = \frac{p}{q}$. Then

$$y = x - (x - y) = \frac{a}{b} - \frac{p}{q} = \frac{aq - bp}{bq}.$$

By properties of integer multiplication and subtraction, $aq - bp \in \mathbb{Z}$ and $bq \in \mathbb{Z}$, and since b, q are both nonzero $bq \neq 0$. Thus, by definition of $\mathbb{Q}$, $y \in \mathbb{Q}$, namely, r . Thus, we proved $\neg p \rightarrow (r \wedge \neg r)$ and so we conclude that p holds.

- (b) If a, b, c are integers and $a^2 + b^2 = c^2$, then at least one of a and b is even.

We prove two helper claims (lemmas) before the proof.

Lemma 1: For any integer x , if x^2 is even, then so is x .

Proof of Lemma 1: Let x be an arbitrary integer and assume (towards proof by contrapositive) that x is not even (hence, odd). WTS that x^2 is also not even. By assumption, $x = 2k + 1$ for some integer k . Squaring: $x^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, and by properties of integer addition and multiplication $2k^2 + 2k \in \mathbb{Z}$, so x^2 is odd, hence not even.

Lemma 2: For any odd integer x , $x^2 \bmod 4 = 1$.

Proof of Lemma 2: Let x be an arbitrary odd integer. By definition, this means there is an integer, call it k , such that $x = 2k + 1$. Squaring: $x^2 = 4k^2 + 4k + 1 = 4(k^2 + k) + 1$. Since $k^2 + k \in \mathbb{Z}$, $x^2 \bmod 4 = 1$.

Proof: Let a, b, c be arbitrary integers. To show: $a^2 + b^2 = c^2 \rightarrow (E(a) \vee E(b))$. Towards a contradiction, consider

$$p = (a^2 + b^2 = c^2) \rightarrow (E(a) \vee E(b)) \quad r = (c^2 \bmod 4 = 0)$$

To show: $\neg p \rightarrow (r \wedge \neg r)$. Assume $\neg p$, that is $a^2 + b^2 = c^2$ and a and b are both not even. By properties of even/odd, a and b are both odd. Thus, by Lemma 2, $a^2 \bmod 4 = b^2 \bmod 4 = 1$. By modular arithmetic, $c^2 \bmod 4 = (a^2 + b^2) \bmod 4 = (1 + 1) \bmod 4 = 2$. Since $2 \neq 0$, this proves $\neg r$. To show: r . Since $c^2 \bmod 4 = 2$, there is an integer, call it q , such that $c^2 = 4q + 2 = 2(2q + 1)$. Thus, c^2 is even. Therefore, applying Lemma 1, c is even. By definition, this gives an integer k such that $c = 2k$. Thus, $c^2 = 4k^2 = 4k^2 + 0$ so $c^2 \bmod 4 = 0$, namely r . Thus, we proved $\neg p \rightarrow (r \wedge \neg r)$ and so we conclude that p holds.

(c) For any integer n , $n^2 + 5$ is not divisible by 4.

Let n be an arbitrary integer. Towards proof by cases, notice that $(n \bmod 4 = 0) \vee (n \bmod 4 = 1) \vee (n \bmod 4 = 2) \vee (n \bmod 4 = 3)$

- **Case 1 to show:** $(n \bmod 4 = 0) \rightarrow (n^2 + 5 \text{ is not divisible by } 4)$. Assume $(n \bmod 4 = 0)$. Then, by modular arithmetic,

$$(n^2 + 5) \bmod 4 = ((n \bmod 4)^2 + 5 \bmod 4) \bmod 4 = (0^2 + 1) \bmod 4 = 1 \bmod 4.$$

Since $(n^2 + 5) \bmod 4 \neq 0$, $n^2 + 5$ is not divisible by 4, as required.

- **Case 2 to show:** $(n \bmod 4 = 1) \rightarrow (n^2 + 5 \text{ is not divisible by } 4)$. Assume $(n \bmod 4 = 1)$. Then, by modular arithmetic,

$$(n^2 + 5) \bmod 4 = ((n \bmod 4)^2 + 5 \bmod 4) \bmod 4 = (1^2 + 1) \bmod 4 = 2 \bmod 4.$$

Since $(n^2 + 5) \bmod 4 \neq 0$, $n^2 + 5$ is not divisible by 4, as required.

- **Case 3 to show:** $(n \bmod 4 = 2) \rightarrow (n^2 + 5 \text{ is not divisible by } 4)$. Assume $(n \bmod 4 = 2)$. Then, by modular arithmetic,

$$(n^2 + 5) \bmod 4 = ((n \bmod 4)^2 + 5 \bmod 4) \bmod 4 = (2^2 + 1) \bmod 4 = 5 \bmod 4 = 1$$

Since $(n^2 + 5) \bmod 4 \neq 0$, $n^2 + 5$ is not divisible by 4, as required.

- **Case 4 to show:** $(n \bmod 4 = 3) \rightarrow (n^2 + 5 \text{ is not divisible by } 4)$.

$$(n^2 + 5) \bmod 4 = ((n \bmod 4)^2 + 5 \bmod 4) \bmod 4 = (3^2 + 1) \bmod 4 = 10 \bmod 4 = 2$$

Since $(n^2 + 5) \bmod 4 \neq 0$, $n^2 + 5$ is not divisible by 4, as required.

The proof by cases is now complete for the arbitrary integer n .

- (d) For all positive integers a, b, c , if $a \nmid bc$ then $a \nmid b$. (The notation $a \nmid b$ means “ a does not divide b ”).

Let a, b, c be arbitrary positive integers. Assume, towards a proof by contrapositive, that $a \mid b$. To show: $a \mid bc$. By definition of divisibility, there is an integer k such that

$$b = ak.$$

Multiplying both sides by c ,

$$bc = (ak)c = a(kc).$$

By properties of integer multiplication, $kc \in \mathbb{Z}$ and so it witnesses $a \mid bc$, as required.

Bonus: Watch the video <https://www.youtube.com/watch?v=MhJN9sByRS0> and write down the statement and one or both of the proofs described using the notation, definitions, and proof strategies from CSE 20.

11. Sets Prove or disprove each of the following statements.

- (a) For all sets A and B , $\mathcal{P}(A) \cup \mathcal{P}(B) = \mathcal{P}(A \cup B)$.

False. A counterexample would be $A = \{1\}, B = \{2\}$. Then

$$\mathcal{P}(A) \cup \mathcal{P}(B) = \{\emptyset, \{1\}\} \cup \{\emptyset, \{2\}\} = \{\emptyset, \{1\}, \{2\}\}$$

but

$$\mathcal{P}(A \cup B) = \mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$$

These sets are not equal because they disagree about membership of $\{1, 2\}$.

- (b) For all sets A and B , $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$.

True. Proof: let A, B be arbitrary sets. We WTS that $\mathcal{P}(A) \cap \mathcal{P}(B) \subseteq \mathcal{P}(A \cap B)$ and $\mathcal{P}(A \cap B) \subseteq \mathcal{P}(A) \cap \mathcal{P}(B)$.

- (c) For the first subset inclusion, consider arbitrary $X \in \mathcal{P}(A) \cap \mathcal{P}(B)$. By definition of intersection, $X \in \mathcal{P}(A)$ and $X \in \mathcal{P}(B)$. By definition of power set, this means that $X \subseteq A$ and $X \subseteq B$. WTS that $X \subseteq A \cap B$: let $x \in X$. Since $X \subseteq A$, $x \in A$. Since $X \subseteq B$, $x \in B$. Thus, $x \in A$ and $x \in B$ so $x \in A \cap B$. Since x was arbitrary, $\forall x(x \in X \rightarrow x \in A \cap B)$ so $X \subseteq A \cap B$. Therefore, by definition of power set $X \in \mathcal{P}(A \cap B)$, as required.

- (d) For the second subset inclusion, consider arbitrary $X \in \mathcal{P}(A \cap B)$. By definition of power set, $X \subseteq (A \cap B)$. WTS that $X \subseteq A$ and $X \subseteq B$. Let $x \in X$. Then by definition of subsets, $x \in A \cap B$. By definition of intersection, $x \in A$ and $x \in B$. Thus $X \subseteq A$ and $X \subseteq B$ and so by definition of power set, $X \in \mathcal{P}(A)$ and $X \in \mathcal{P}(B)$. By definition of intersection, $X \in \mathcal{P}(A) \cap \mathcal{P}(B)$, as required.

- (e) For any sets A, B, C, D , if the Cartesian products $A \times B$ and $C \times D$ are disjoint then either A and C are disjoint or B and D are disjoint (or both).

True. Proof: Let A, B, C, D be arbitrary sets. Towards a proof by contrapositive, we assume that $A \cap C \neq \emptyset$ and $B \cap D \neq \emptyset$ and WTS that $(A \times B) \cap (C \times D) \neq \emptyset$. Since $A \cap C \neq \emptyset$, let $x \in A \cap C$. Since $B \cap D \neq \emptyset$, let $y \in B \cap D$. Let's consider (x, y) . By definition of intersection, since $x \in A \cap C$ and $y \in B \cap D$, $x \in A$ and $y \in B$. Thus, by definition of Cartesian product, $(x, y) \in A \times B$. Similarly, $(x, y) \in C \times D$. Thus, by definition of intersection, $(x, y) \in (A \times B) \cap (C \times D)$. In particular, this means that $(A \times B) \cap (C \times D) \neq \emptyset$, as required.

- (f) There are sets A, B such that $A \in B$ and $A \subseteq B$.

True. Proof: Consider the example $A = \{1, 2\}$ and $B = \{1, 2, \{1, 2\}\}$. Then $A \in B$ because $\{\{1, 2\}\}$ shows up in the list of elements of B . Moreover, since each of the elements of A (the numbers 1 and 2) are also elements of B (they each show up in B 's list of elements), A is a subset of B .

- (g) For all sets A, B, C : $A \cap B = \emptyset$ and $B \cap C = \emptyset$ if and only if $(A \cap B) \cap C = \emptyset$.

False. Proof: Consider the counterexample $A = \{1, 2\}$ and $B = \{2, 3\}$ and $C = \{3\}$. Then it is not the case that “ $A \cap B = \emptyset$ and $B \cap C = \emptyset$ ” because $A \cap B = \{2\}$. However, it is the case that $(A \cap B) \cap C = \emptyset$ because

$$(A \cap B) \cap C = (\{1, 2\} \cap \{2, 3\}) \cap \{3\} = \{2\} \cap \{3\} = \emptyset.$$

The biconditional statement is false because one of its arguments is false while the other is true.

12. Induction and Recursion Prove the following statements:

- (a) $\forall s \in S (rnalen(s) = basecount(s, A) + basecount(s, C) + basecount(s, U) + basecount(s, G))$
where S RNA strands and the functions *rnalen* and *basecount* are define recursively as:

$$\begin{array}{lll} & & rnalen : S \rightarrow \mathbb{Z}^+ \\ \text{Basis Step:} & \text{If } b \in B \text{ then} & rnalen(b) = 1 \\ \text{Recursive Step:} & \text{If } s \in S \text{ and } b \in B, \text{ then} & rnalen(sb) = 1 + rnalen(s) \\ & & basecount : S \times B \rightarrow \mathbb{N} \\ \text{Basis Step:} & \text{If } b_1 \in B, b_2 \in B & basecount(b_1, b_2) = \begin{cases} 1 & \text{when } b_1 = b_2 \\ 0 & \text{when } b_1 \neq b_2 \end{cases} \\ \text{Recursive Step:} & \text{If } s \in S, b_1 \in B, b_2 \in B & basecount(sb_1, b_2) = \begin{cases} 1 + basecount(s, b_2) & \text{when } b_1 = b_2 \\ basecount(s, b_2) & \text{when } b_1 \neq b_2 \end{cases} \end{array}$$

We proceed by structural induction:

- Basis step: We consider the four cases of $b \in \{\mathbf{A}, \mathbf{C}, \mathbf{U}, \mathbf{G}\}$, in each to show is that $rnalen(b) = basecount(b, \mathbf{A}) + basecount(b, \mathbf{C}) + basecount(b, \mathbf{U}) + basecount(b, \mathbf{G})$.

$$rnalen(\mathbf{A}) = 1 \quad \text{by the basis step in the recursive definition of } rnalen$$

$$basecount(\mathbf{A}, \mathbf{A}) + basecount(\mathbf{A}, \mathbf{C}) + basecount(\mathbf{A}, \mathbf{U}) + basecount(\mathbf{A}, \mathbf{G}) = 1 + 0 + 0 + 0 = 1$$

by the basis step in the recursive definition of *basecount*, where the first term is the case where $b_1 = b_2 = \mathbf{A}$ and the other three terms are the case where $b_1 \neq b_2$. Thus, $rnalen(\mathbf{A}) = 1 = basecount(\mathbf{A}, \mathbf{A}) + basecount(\mathbf{A}, \mathbf{C}) + basecount(\mathbf{A}, \mathbf{U}) + basecount(\mathbf{A}, \mathbf{G})$.

Similarly,

$$rnalen(\mathbf{C}) = 1 \quad \text{by the basis step in the recursive definition of } rnalen$$

$$basecount(\mathbf{C}, \mathbf{A}) + basecount(\mathbf{C}, \mathbf{C}) + basecount(\mathbf{C}, \mathbf{U}) + basecount(\mathbf{C}, \mathbf{G}) = 0 + 1 + 0 + 0 = 1$$

by the basis step in the recursive definition of *basecount*, where the second term is the case where $b_1 = b_2 = \mathbf{C}$ and the other three terms are the case where $b_1 \neq b_2$. Thus, $rnalen(\mathbf{C}) = 1 = basecount(\mathbf{C}, \mathbf{A}) + basecount(\mathbf{C}, \mathbf{C}) + basecount(\mathbf{C}, \mathbf{U}) + basecount(\mathbf{C}, \mathbf{G})$;

$$rnalen(\mathbf{U}) = 1 \quad \text{by the basis step in the recursive definition of } rnalen$$

$$basecount(\mathbf{U}, \mathbf{A}) + basecount(\mathbf{U}, \mathbf{C}) + basecount(\mathbf{U}, \mathbf{U}) + basecount(\mathbf{U}, \mathbf{G}) = 0 + 0 + 1 + 0 = 1$$

by the basis step in the recursive definition of *basecount*, where the third term is the case where $b_1 = b_2 = \mathbf{U}$ and the other three terms are the case where $b_1 \neq b_2$. Thus, $rnalen(\mathbf{U}) = 1 = basecount(\mathbf{U}, \mathbf{A}) + basecount(\mathbf{U}, \mathbf{C}) + basecount(\mathbf{U}, \mathbf{U}) + basecount(\mathbf{U}, \mathbf{G})$;

$$rnalen(\mathbf{G}) = 1 \quad \text{by the basis step in the recursive definition of } rnalen$$

$$basecount(\mathbf{G}, \mathbf{A}) + basecount(\mathbf{G}, \mathbf{C}) + basecount(\mathbf{G}, \mathbf{U}) + basecount(\mathbf{G}, \mathbf{G}) = 0 + 0 + 0 + 1 = 1$$

by the basis step in the recursive definition of *basecount*, where the last term is the case where $b_1 = b_2 = \mathbf{G}$ and the other three terms are the case where $b_1 \neq b_2$. Thus, $rnalen(\mathbf{G}) = 1 = basecount(\mathbf{G}, \mathbf{A}) + basecount(\mathbf{G}, \mathbf{C}) + basecount(\mathbf{G}, \mathbf{U}) + basecount(\mathbf{G}, \mathbf{G})$. The basis step is now complete.

- Recursive step: Consider an arbitrary strand s and an arbitrary base b . Assume, as the **induction hypothesis** that

$$rnalen(s) = basecount(s, \mathbf{A}) + basecount(s, \mathbf{C}) + basecount(s, \mathbf{U}) + basecount(s, \mathbf{G})$$

(continued next page)

- (Recursive step, continued). We need to show that

$$r_{nalen}(sb) = \text{basecount}(sb, \mathbf{A}) + \text{basecount}(sb, \mathbf{C}) + \text{basecount}(sb, \mathbf{U}) + \text{basecount}(sb, \mathbf{G})$$

We consider the four cases of $b \in \{\mathbf{A}, \mathbf{C}, \mathbf{U}, \mathbf{G}\}$. In each case, the LHS of the to show is

$$r_{nalen}(sb) = 1 + r_{nalen}(s) \quad \text{by the recursive step in the definition of } r_{nalen}$$

For the RHS, when $b = \mathbf{A}$:

$$\begin{aligned} RHS &= \text{basecount}(s\mathbf{A}, \mathbf{A}) + \text{basecount}(s\mathbf{A}, \mathbf{C}) + \text{basecount}(s\mathbf{A}, \mathbf{U}) + \text{basecount}(s\mathbf{A}, \mathbf{G}) \\ &= (1 + \text{basecount}(s, \mathbf{A})) + \text{basecount}(s, \mathbf{C}) + \text{basecount}(s, \mathbf{U}) + \text{basecount}(s, \mathbf{G}) \\ &= 1 + (\text{basecount}(s, \mathbf{A}) + \text{basecount}(s, \mathbf{C}) + \text{basecount}(s, \mathbf{U}) + \text{basecount}(s, \mathbf{G})) \\ &= 1 + r_{nalen}(s) \quad \text{by the induction hypothesis} \end{aligned}$$

(where the first equation listed is by the recursive step in the definition of basecount, the first term from the case $b_1 = b_2 = \mathbf{A}$ and the rest of the terms from the case $b_1 \neq b_2$). Thus, $LHS = r_{nalen}(s\mathbf{A}) = 1 + r_{nalen}(s) = RHS$.

Similarly, when $b = \mathbf{C}$:

$$\begin{aligned} RHS &= \text{basecount}(s\mathbf{C}, \mathbf{A}) + \text{basecount}(s\mathbf{C}, \mathbf{C}) + \text{basecount}(s\mathbf{C}, \mathbf{U}) + \text{basecount}(s\mathbf{C}, \mathbf{G}) \\ &= \text{basecount}(s, \mathbf{A}) + (1 + \text{basecount}(s, \mathbf{C})) + \text{basecount}(s, \mathbf{U}) + \text{basecount}(s, \mathbf{G}) \\ &= 1 + r_{nalen}(s) \quad \text{by the induction hypothesis} \end{aligned}$$

(where the first equation listed is by the recursive step in the definition of basecount, the second term from the case $b_1 = b_2 = \mathbf{C}$ and the rest of the terms from the case $b_1 \neq b_2$). Thus, $LHS = r_{nalen}(s\mathbf{C}) = 1 + r_{nalen}(s) = RHS$. When $b = \mathbf{U}$:

$$\begin{aligned} RHS &= \text{basecount}(s\mathbf{U}, \mathbf{A}) + \text{basecount}(s\mathbf{U}, \mathbf{C}) + \text{basecount}(s\mathbf{U}, \mathbf{U}) + \text{basecount}(s\mathbf{U}, \mathbf{G}) \\ &= \text{basecount}(s, \mathbf{A}) + \text{basecount}(s, \mathbf{C}) + (1 + \text{basecount}(s, \mathbf{U})) + \text{basecount}(s, \mathbf{G}) \\ &= 1 + r_{nalen}(s) \quad \text{by the induction hypothesis} \end{aligned}$$

(where the first equation listed is by the recursive step in the definition of basecount, the third term from the case $b_1 = b_2 = \mathbf{U}$ and the rest of the terms from the case $b_1 \neq b_2$). Thus, $LHS = r_{nalen}(s\mathbf{U}) = 1 + r_{nalen}(s) = RHS$. Finally, for $b = \mathbf{G}$:

$$\begin{aligned} RHS &= \text{basecount}(s\mathbf{G}, \mathbf{A}) + \text{basecount}(s\mathbf{G}, \mathbf{C}) + \text{basecount}(s\mathbf{G}, \mathbf{U}) + \text{basecount}(s\mathbf{G}, \mathbf{G}) \\ &= \text{basecount}(s, \mathbf{A}) + \text{basecount}(s, \mathbf{C}) + \text{basecount}(s, \mathbf{U}) + (1 + \text{basecount}(s, \mathbf{G})) \\ &= 1 + r_{nalen}(s) \quad \text{by the induction hypothesis} \end{aligned}$$

(where the first equation listed is by the recursive step in the definition of basecount, the last term from the case $b_1 = b_2 = \mathbf{G}$ and the rest of the terms from the case $b_1 \neq b_2$). Thus, $LHS = r_{nalen}(s\mathbf{G}) = 1 + r_{nalen}(s) = RHS$.

- (b) $\forall l_1 \in L \forall l_2 \in L (sum(concat(l_1, l_2)) = sum(l_1) + sum(l_2))$, where the function *concat* is defined as:

$$\begin{array}{lll} & concat : L \times L & \rightarrow L \\ \text{Basis Step:} & \text{If } l \in L & concat([], l) = l \\ \text{Recursive Step:} & \text{If } l, l' \in L \text{ and } n \in \mathbb{N}, \text{ then} & concat((n, l), l') = (n, concat(l, l')) \end{array}$$

and the function $sum : L \rightarrow \mathbb{N}$ that sums all the elements of a list and is defined by:

$$\begin{array}{lll} & sum : L & \rightarrow \mathbb{N} \\ \text{Basis Step:} & sum([]) & = 0 \\ \text{Recursive Step:} & \text{If } l \in L, n \in \mathbb{N} & sum((n, l)) = n + sum(l) \end{array}$$

We proceed by structural induction on l_1 by definition of L .

- Basis step: To show is that $\forall l_2 \in L (sum(concat([], l_2)) = sum([]) + sum(l_2))$. We proceed by universal generalization and let l_2 be an arbitrary element of L . By definition of *concat*, we can rewrite the To Show as $sum(l_2) = sum([]) + sum(l_2)$. By definition of *sum*, we can rewrite the right-hand side as $0 + sum(l_2)$. This shows that our goal can be rewritten to $sum(l_2) = sum(l_2)$, which is true.
- Recursive step: To show is that $\forall l_2 \in L (sum(concat((n, l'_1), l_2)) = sum((n, l'_1)) + sum(l_2))$, where $n \in \mathbb{N}$ and $l'_1 \in L$. We assume as the inductive hypothesis that $\forall l_2 \in L (sum(concat(l'_1, l_2)) = sum(l'_1) + sum(l_2))$.

We proceed by universal generalization and assume that l_2 is an arbitrary element of L . Applying the definition of *concat*, we can rewrite To Show as

$$sum((n, concat(l'_1, l_2))) = sum((n, l'_1)) + sum(l_2)$$

. We can further apply the definition of *sum* on both sides to rewrite To Show as:

$$n + sum(concat(l'_1, l_2)) = n + sum(l'_1) + sum(l_2)$$

. By the rules of arithmetic on $+$, this is true when

$$sum(concat(l'_1, l_2)) = sum(l'_1) + sum(l_2)$$

. Since $l_2 \in L$, we can apply the inductive hypothesis, whose body exactly matches this goal.

13. Induction and Recursion At time 0, a particle resides at the point 0 on the real line. Within 1 second, it divides into 2 particles that fly in opposite directions and stop at distance 1 from the original particle. Within the next second, each of these particles again divides into 2 particles flying in opposite directions and stopping at distance 1 from the point of division, and so on. Whenever particles meet they annihilate (leaving nothing behind). How many particles will there be at time $2^{11} - 1$? You do not need

to justify your answer. Hint: Derive a formula for the number/ locations of particles at time $2^n - 1$ for arbitrary positive integer n , prove the formula using induction, and apply it when $n = 11$.

2^{11} particles

Proof: We will prove by mathematical induction that,

For each positive integer n , a particle that started at time 0 in position 0 has the property that between time 0 and time $2^n - 1$, it and all of its descendants have stayed in the interval $[-2^n + 1, 2^n - 1]$ and at time $2^n - 1$, its descendants are located exactly on each odd location in $[-2^n + 1, 2^n - 1]$ (inclusive).

How to use this lemma? Substituting $n = 11$, this lemma says that at time $2^{11} - 1$, the descendants of the original particle are located at odd each location in the interval $[-2^{11} + 1, 2^{11} - 1]$ (inclusive). Therefore, the number of particles at this time is the number of odd integers in the interval. There are $\lceil \frac{2^{11}-1}{2} \rceil = \frac{2^{11}}{2} = 2^{10}$ many such negative odd integers, and $\lceil \frac{2^{11}-1}{2} \rceil = \frac{2^{11}}{2} = 2^{10}$ many such positive odd integers, so a total of $2^{10} + 2^{10} = 2^{11}$ particles.

To prove the lemma, we proceed by mathematical induction.

- **Basis step** For $n = 1$, we WTS that between time 0 and time $2^1 - 1$, the particle and all its descendants have stayed in the interval $[-2^1 + 1, 2^1 - 1]$, and at time $2^1 - 1$, there are particles at each odd location this interval. Plugging in $n = 1$, we see that the interval is $[-1, 1]$ and we WTS that at time 1 there are particles exactly at the endpoints of this interval (these are the odd locations). Applying the definition of the process, at time 1, the original particle divides into two and the two new particles reach distance 1 from the original. This means that the only locations at which particles resided in this interval are 0, -1 , 1 (all within the required interval) and that at time 1, the particles are at -1 and 1 , as required.

(continued next page)

- **Induction step** Let k be an arbitrary positive integer. Assume, as the **Induction Hypothesis (IH)**, that

between time 0 and time $2^k - 1$, the particle and all of its descendants have stayed in the region on the number line $[-2^k + 1, 2^k - 1]$ and at time $2^k - 1$, its descendants are located exactly on each odd location in $[-2^k + 1, 2^k - 1]$ (inclusive).

We WTS that

between time 0 and time $2^{k+1} - 1$, the particle and all of its descendants have stayed in the region on the number line $[-2^{k+1} + 1, 2^{k+1} - 1]$ and at time $2^{k+1} - 1$, its descendants are located exactly on each odd location in $[-2^{k+1} + 1, 2^{k+1} - 1]$ (inclusive).

We'll consider the progress of the experiment through time. At time step 2^k , one unit of time has elapsed since the situation described in the IH. By the setup of the experiment, during this unit of time, each particle splits into two and these travel one unit in each axis direction. Since (by the IH) at time $2^k - 1$, there are particles at each odd location in the interval $[-2^k + 1, 2^k - 1]$, each of these particles are two units apart. Thus, the particles originating from successive odd locations in time $2^k - 1$ will meet and annihilate at time 2^k . The only particles that will remain at time 2^k are (1) the particle that originates at time $2^k - 1$ from location $-2^k + 1$ and goes left and (2) the particle that originates at time $2^k - 1$ from location $2^k - 1$ and goes right. Thus, at 2^k there are two particles, one at -2^k and the other at 2^k . For definiteness, let's call the particle at position -2^k , p_L , and the one at position 2^k , p_R . Recalibrating the experiment with position 2^k as the new "origin" and particle p_R as our original particle, the IH guarantees that between the current time (2^k) and $2^k - 1$ steps in the future, p_R will divide into particles such that all of its descendants will stay in the interval $["origin" - 2^k + 1, "origin" + 2^k - 1]$ and its descendants will end up exactly on each odd location in $["origin" - 2^k + 1, "origin" + 2^k - 1]$ (inclusive). That is, at time $2^k + 2^k - 1 = 2^{k+1} - 1$, the process starting p_R will have generated particles at all odd locations in $[0, 2^{k+1} - 1]$. Symmetrically, the IH guarantees that at time $2^{k+1} - 1$, all descendants of the particle p_L will be at the odd locations in $[-2^{k+1} + 1, 0]$, and will never have gone beyond that interval at any time since time $2^k - 1$. Since the intervals in which the descendants of p_L and p_R land between these timestamps only overlap at 0, the particles generated by these two experiments never collide and so the two experiments run independently. Thus, at time $2^{k+1} - 1$, there are $2^k + 2^k = 2^{k+1}$ many particles, occupying each odd location in the interval $[-2^{k+1} + 1, 0] \cup [0, 2^{k+1} - 1] = [-2^{k+1} + 1, 2^{k+1} - 1]$, and these particles (and their predecessors) have not left this interval at any point up to this time step. In particular, this means that the lemma is proved.

- 14. Induction and Recursion** Prove that every positive integer has a base 3 expansion.
Hint: use strong induction.

By definition of base expansion, we need to prove that for every positive integer n , there is a positive integer k and nonnegative integers $a_0, a_1, \dots, a_{k-1}$ such that each $a_i \in \{0, 1, 2\}$, $a_{k-1} \neq 0$, and

$$n = \sum_{i=0}^{k-1} a_i 3^i$$

We proceed by strong induction on $n \geq 0$.

- Basis steps (using the terminology for strong induction: we choose $b = 1, j = 1$): For the positive integer 1, we have witnesses $k = 1, a_0 = 1$ and since $1 \in \{0, 1, 2\}$, $1 \neq 0$, and $1 = \sum_{i=0}^0 a_i 3^i$, the claim is proved. For the positive integer 2, we have witnesses $k = 1, a_0 = 2$ and since $2 \in \{0, 1, 2\}$, $2 \neq 0$, and $2 = \sum_{i=0}^0 a_i 3^i$, the claim is proved.
- Recursive step: Consider an arbitrary positive integer n greater than or equal to 2. Assume, as the strong induction hypothesis that each positive number less than or equal to n has a base 3 expansion. We need to show that $n + 1$ has a base 3 expansion. To build the required witnesses, consider the integer $c = (n + 1) \text{ div } 3$. By properties of integers, since $n + 1 \geq 3$ (because $n \geq 2$), $1 \leq c \leq n$. Thus, the strong induction hypothesis applies to c and there is a positive integer k_c and nonnegative integers $x_0, x_1, \dots, x_{k_c-1}$ such that each $x_i \in \{0, 1, 2\}$, $x_{k_c-1} \neq 0$, and

$$c = \sum_{i=0}^{k_c-1} x_i 3^i$$

Define $k = k_c + 1$, $a_i = x_{i-1}$ for each i from 1 to k , and $a_0 = (n + 1) \text{ mod } 3$. Then k is positive (because k_c is), each $a_i \in \{0, 1, 2\}$ (because all the x_i are in this set, and the remainder upon division by 3 of any positive integer is also in this set), $a_{k-1} = a_{k_c+1-1} = x_{k_c-1} \neq 0$ (by choice of k and definition of x_i s). It remains to prove that the sum gives the right value:

$$\begin{aligned} \sum_{i=0}^{k-1} a_i 3^i &= \left(\sum_{i=1}^{k-1} a_i 3^i \right) + a_0 3^0 = 3 \left(\sum_{i=1}^{(k_c+1)-1} x_{i-1} 3^{i-1} \right) + a_0 3^0 = 3 \left(\sum_{j=0}^{k_c-1} x_j 3^j \right) + a_0 \\ &= 3c + a_0 = 3((n + 1) \text{ div } 3) + ((n + 1) \text{ mod } 3) = n + 1 \end{aligned}$$

as required.

Since the existence of base 3 expansions was proved for all positive integers (for 1, 2 in the basis cases, and all integers greater than or equal to 3 in the recursive step), the proof by strong induction is complete.

15. Functions & Cardinalities of sets Prove each of the following claims.

- (a) For the “function” $f : \mathbb{Z} \rightarrow \{0, 1, 2, 3\}$ given by $f(x) = x \text{ mod } 5$, it is not the case that every element of the domain maps to exactly one element of the codomain

(that is, it is not a well-defined function).

This function is not well defined because $f(4)$ is not in the codomain even though 4 is in the domain: $f(4) = 4 \bmod 5 = 4 \notin \{0, 1, 2, 3\}$.

- (b) The “function” $f : \mathcal{P}(\mathbb{Z}^+) \rightarrow \mathbb{Z}$ given by

$$f(A) = \text{the maximum element in } A$$

is not well-defined.

This function is not well defined because $f(\mathbb{Z}^+)$ is not well-defined even though $\mathbb{Z}^+$ is in the domain: the set $\mathbb{Z}^+$ doesn't have a maximum element.

- (c) There is a one-to-one function with domain $\{a, b, c\}$ and codomain $\mathbb{R}$.

Define the function $f : \{a, b, c\} \rightarrow \mathbb{R}$ by the piecewise definition

$$f(a) = 1, f(b) = 2, f(c) = 3$$

This function is well-defined because it maps every domain element to a unique output, and all outputs are in the specified codomain. Moreover, it is one-to-one: each of the three distinct domain elements have different images from one another.

- (d) There is an onto function with domain R and codomain $\{\pi, \frac{1}{17}\}$, where R is the set of user ratings in a database with 5 movies.

Define the function $f : R \rightarrow \{\pi, \frac{1}{17}\}$ by the piecewise definition

$$f(w) = \begin{cases} \pi & \text{if } w = (0, 0, 0, 0, 0) \\ \frac{1}{17} & \text{otherwise} \end{cases}$$

This function is well-defined because it maps every ratings 5-tuple to a unique output, and all outputs are in the specified codomain. Moreover, it is onto: we need to confirm that each element in the codomain has a preimage. Consider, for example

$$f((0, 0, 0, 0, 0)) = \pi \quad f((1, 1, 1, 1, 1)) = \frac{1}{17}.$$

- (e) The Cartesian product $\mathbb{Z}^+ \times \{a, b, c\}$ is countable.

We rewrite

$$\mathbb{Z}^+ \times \{a, b, c\} = (\mathbb{Z}^+ \times \{a\}) \cup (\mathbb{Z}^+ \times \{b\}) \cup (\mathbb{Z}^+ \times \{c\})$$

Using the function $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+ \times \{a\}$ given by $f(n) = (n, a)$, we can prove that $\mathbb{Z}^+ \times \{a\}$ is countably infinite. Similarly, $\mathbb{Z}^+ \times \{b\}$ and $\mathbb{Z}^+ \times \{c\}$ are countably infinite. By Theorem 1 on page 174 in the book, $(\mathbb{Z}^+ \times \{a\}) \cup (\mathbb{Z}^+ \times \{b\})$ is countable because it is the union of two countably infinite (hence countable) sets. Applying this theorem again when taking the union of this resulting set with the set $(\mathbb{Z}^+ \times \{c\})$, we see that the set is countable.

- (f) The interval of real numbers $\{x \in \mathbb{R} \mid 5 \leq x \leq 8\}$ is uncountable. *Hint: you may use the fact that the unit interval $\{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$ is uncountable.*

For brevity, we use the interval notation $[5, 8] = \{x \in \mathbb{R} \mid 5 \leq x \leq 8\}$ and $[0, 1] = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$. Define the function $f : [0, 1] \rightarrow [5, 8]$ by $f(x) = 5 + 3x$. Then f is a bijection:

- Well-defined? for each x in $[0, 1]$, $0 \leq x \leq 1$ so $0 \leq 3x \leq 3$ and $5 \leq 5 + 3x \leq 8$. Thus, $f(x) \in [5, 8]$, as required.
- Invertible? Consider the function $g : [5, 8] \rightarrow [0, 1]$ defined by $g(x) = \frac{x-5}{3}$. WTS that for each $x \in [0, 1]$, $g(f(x)) = x$ and for each $x \in [5, 8]$, $f(g(x)) = x$. Let $x \in [0, 1]$. Then

$$g(f(x)) = g(5 + 3x) = \frac{(5 + 3x) - 5}{3} = \frac{3x}{3} = x,$$

as required. Similarly, let $x \in [5, 8]$. Then

$$f(g(x)) = f\left(\frac{x-5}{3}\right) = 5 + 3 \cdot \frac{x-5}{3} = 5 + (x-5) = x,$$

as required. Thus, f is invertible (with inverse g) and so is a bijection.